

Appendix D. Simulation workflow and ODE solver specifications

The simulation logic is formalized in Algorithm 1. The tripartite replicator dynamics system was solved using MATLAB’s built-in `ode45` solver, which implements a variable-step Runge-Kutta (4,5) method. To ensure numerical stability, we enforced a relative tolerance of `RelTol` = 1e-6 and an absolute tolerance of `AbsTol` = 1e-8 over the integration time span $t \in [0, 100]$. The algorithm details the execution of the primary scenarios and the multi-origin phase space analysis.

Algorithm 1 Simulation workflow for tripartite evolutionary dynamics across multiple scenarios

Require: Global baseline parameters: $C_{p1}, C_{p2}, u_1, E_m, E_u, C_b, f_{b1}, f_{b2}, R_b, C_u, f_u, R_{p1}, R_{p2}$

Require: ODE Solver settings: Time span $T = [0, 100]$, $RelTol = 10^{-6}$, $AbsTol = 10^{-8}$

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1: Define scenarios  $\mathcal{S} = \{1 : \text{Scenario I}, 2 : \text{Scenario II}, 3 : \text{Scenario III}, 4 : \text{Robustness Check}\}$ 
2: Define Replicator Dynamics function  $F(t, \mathbf{x})$ , where  $\mathbf{x} = [\alpha, \beta, \gamma]^T$ :
3:    $\dot{\alpha} = \alpha(1 - \alpha) [f_p + \gamma(1 - \beta)R_{p2} - C_{p2}]$ 
4:    $\dot{\beta} = \beta(1 - \beta) [E_m + C_b + \gamma R_b + (\gamma + \alpha u_2(1 - \gamma))f_{b1} + \alpha u_2 f_{b2} - (1 - u_1)(1 - \alpha u_2)E_u]$ 
5:    $\dot{\gamma} = \gamma(1 - \gamma) [(1 - u_2\alpha)(1 - \beta)f_u - C_u]$ 
6: for each scenario  $\in \mathcal{S}$  do
7:   Switch scenario do
8:     Case 1:  $f_p \leftarrow 30, u_2 \leftarrow 0.4$  ▷ Weak penalty, weak tech (Trap)
9:     Case 2:  $f_p \leftarrow 400, u_2 \leftarrow 0.3$  ▷ Strong penalty, weak tech (Limit Cycle)
10:    Case 3:  $f_p \leftarrow 120, u_2 \leftarrow 0.9$  ▷ Strong penalty, strong tech (Ideal)
11:    Case 4:  $f_p \leftarrow 30, u_2 \leftarrow 0.4$  ▷ Baseline parameters for robustness
12:   End Switch
13:   if scenario  $\leq 3$  then ▷ Execute standard 2D time-evolution analysis
14:     Set initial condition:  $\mathbf{x}_0 \leftarrow [0.3, 0.5, 0.4]^T$ 
15:      $\mathbf{X}(t) \leftarrow \text{ODE45}(F, T, \mathbf{x}_0, RelTol, AbsTol)$ 
16:     Plot 2D evolutionary trajectories:  $\alpha(t), \beta(t), \gamma(t)$  over  $T$ 
17:   else ▷ Execute 3D phase diagram for robustness check
18:     Define initial condition set  $IC \leftarrow \{[0.3, 0.5, 0.4]^T, [0.1, 0.3, 0.2]^T, [0.5, 0.5, 0.5]^T, [0.7, 0.7, 0.6]^T\}$ 
19:     Initialize 3D phase plot
20:     for each  $\mathbf{x}_0 \in IC$  do
21:        $\mathbf{X}(t) \leftarrow \text{ODE45}(F, T, \mathbf{x}_0, RelTol, AbsTol)$ 
22:       Plot 3D trajectory of  $\mathbf{X}(t)$  in  $(\alpha, \beta, \gamma)$  space
23:       Mark starting point  $\mathbf{x}_0$ 
24:     end for
25:     Mark evolutionary stable strategy (ESS) convergence point  $E_2(0, 0, 1)$ 
26:   end if
27: end for

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